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(ISO 9001:2015 Certified)
Accredited with 'A' Grade by NAAC

Department of Information Science & Engineering

MAJOR PROJECT WORK SYNOPSIS

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Project Code:	**53**
Project Title:	**Machine Learning Based Lifecycle Intelligence System for Non Biodegradable Automotive Materials**

Introduction:

Modern automobiles provide increasing usage of non-biodegradable materials such as composite fibers, reinforced plastics, and polymer-based materials. The increased usage of these materials helps in enhancing automobile performance. Nevertheless, these materials prove to be a huge problem for our environment when vehicles approach their end-of-life stage. If these materials are not properly handled during this stage, soil pollution and loss of reusable resources take place.

The rising number of end-of-life vehicles surpasses the intelligence of current waste management practices, which are further inadequate for analyzing waste materials and determining environmental risks. It was suggested in this project that an efficient and intelligent waste management practice would be carried out using a software application focusing on machine learning for analyzing automotive material data.

Objectives of the Project:

1. To analyze lifecycle data of non-biodegradable automotive materials and perform data preprocessing and feature extraction to make the dataset suitable for machine learning models.

2. To implement supervised machine learning algorithms to classify automotive materials into recyclable, reusable, or disposal categories.

3. To evaluate and compare the performance of the implemented models using standard metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

4. To identify the best-performing model and develop a simple lifecycle intelligence system that supports sustainable decision-making in automotive material management.

Problem Statement:

For instance, though in the automobile industry, the use of non-biodegradable materials has increased, the current system for managing the associated wastes does not have intelligent tools for detecting the kind and degree of risk. There are only traditional approaches that are not effective for managing the increased amount of automobile waste data and could potentially cause inappropriate disposal and damage to the environment, making it necessary for the system to incorporate machine learning for the lifecycle intelligent system.

Methodology:

The project starts with the collection of the material used in the automotive industry, available through publicly accessible sources, datasets, and simulation-based lifecycle data. The collected material data is then preprocessed to address missing values, normalize the features, and select the important features regarding the materials.

Additionally, machine learning models will classify the material type and determine the environmental risk level. The performance of these models will be assessed using measures like accuracy, precision, recall, and error rate. After the evaluation, effective models will be selected and improved through the adjustment of their hyperparameters.

The final system will encompass all of the above modules in an integrated form of a unified computer-based system with an easily usable interface.

Significance and Expected Outcomes:

The aim of this proposed project is that, through this intelligent software, it should be possible to analyze non-biodegradable, automotive-related material through all phases of its existence, which in return helps with decision-making for successfully reusing or disposing of such material, all while considering environmental concerns.

The purpose of this project is to present a practical example for machine learning in sustainability-driven decision-making strategies. Besides, this project serves as a basis for further research directed towards intelligent waste management systems.

Conclusion:

In the following project, the machine learning-based intelligent system for the lifecycle management of non-biodegradable materials used in automotive components is discussed. Here, the emphasis is on the potential for sustainability and the mitigation of environmental degradation through the proposed intelligent approach. Accordingly, the proposition addresses the potential for the intelligent solution for the persistence of environmental problems.

References:

IEEE Style Format:

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A. Sharma and P. Verma, "Machine learning approaches for material classification," International Journal of Computer Applications, vol. 180, no. 25, pp. 12–18, Jul. 2020.

International Energy Agency, "End-of-life vehicle management," IEA. [Online]. Available: https://www.iea.org. [Accessed: Jan. 2026].

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Accepted/Rejected:

Suggestion/Corrections if any Specify:

Date:

Signature of the Project Coordinators: